

AI-powered software development

Accelerating construction productivity from code to jobsite

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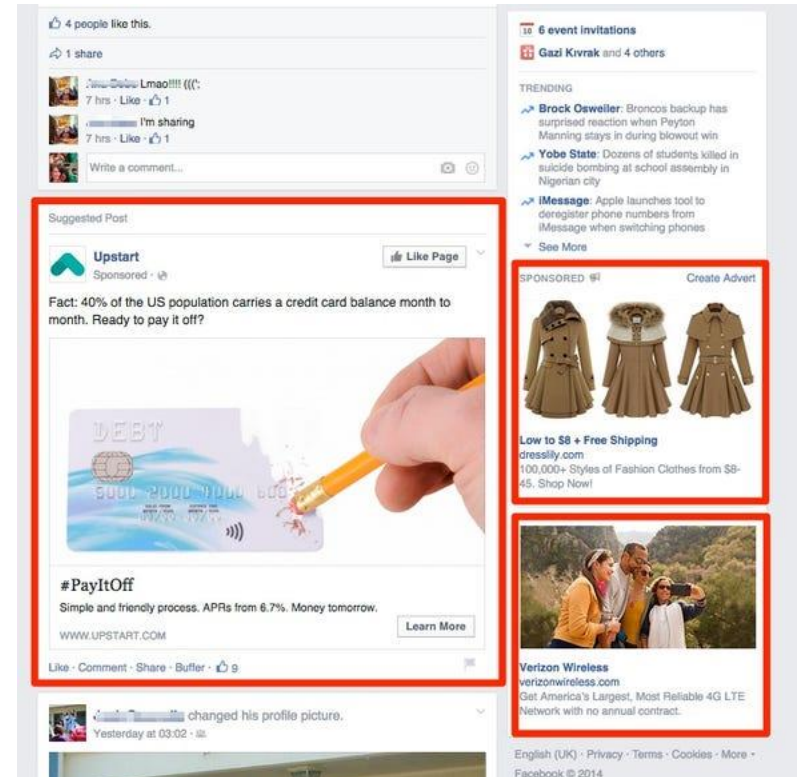


OPENING

We are not yet in an AI-native ConTech world.

There is real progress everywhere:

- better models
- better enterprise tools
- better document understanding



WHERE WE ARE

Most AI in construction is added to existing platforms

The screenshot displays the Procore AI interface integrated into the Procore platform. The top navigation bar includes the Procore logo, a search bar, and a dropdown menu for selecting an app. The main content area is divided into two columns. The left column shows the project details for the Seattle Corridor Railway, including the address 9227 East Forest Street, Austin, TX 73301, and a list of open items categorized by due date (Overdue, Due Today, Due Soon). The right column features the Procore AI chat interface, which includes a chat window with a message from Jim asking to draft an RFI regarding a conflict between structural drawing S-101 and architectural drawing A-205. Below the chat window, there is a section for 'Your Project's Submittal Response Time' with a 'Quick Insight' stating that the project has a better rate of responding to submittals on time than 74% of the projects across the industry. A horizontal bar chart compares the average days to complete submittals for the project, company, and industry. The project bar is at approximately 7.5 days, the company bar is at approximately 7.5 days, and the industry bar is at approximately 2.5 days. The chat interface also includes a 'Current Page' dropdown and a 'Fastest' dropdown.

Menu PROCORE Vertigo Construction Seattle Corridor Railway Search or Ask a Question Ctrl K Apps Select an App ? ? ? ? ? Vertigo CONSTRUCTION

Seattle Corridor Railway COURSE OF CONSTRUCTION 9227 East Forest Street, Austin, TX 73301 87°F 91° | 64° + Project Links Site Logistic Plans Latest Drawings EarthCam Link

Open Items
View all of the open items in this project that require action.
My Open Items All Open Items Group by: Due Date

Overdue (3)
Due Today (2)
Due Soon (10)

Insights
View actionable insights to help increase productivity and reduce risk.

Your Project's Submittal Response Time

Quick Insight
Your project has a better rate of responding to Submittals on time than 74% of the projects across the industry.

Average days to complete Submittals

Category	Average days to complete Submittals
Project	~7.5
Company	~7.5
Industry	~2.5

Procore AI

New Chat

Jim
Draft an RFI regarding the conflict between structural drawing S-101, detail 3/S-101, and architectural drawing A-205, detail 205, concerning the beam penetration.

RFI Creation Agent 3 steps
I've got all the pieces to draft this RFI...

Current Page Fastest

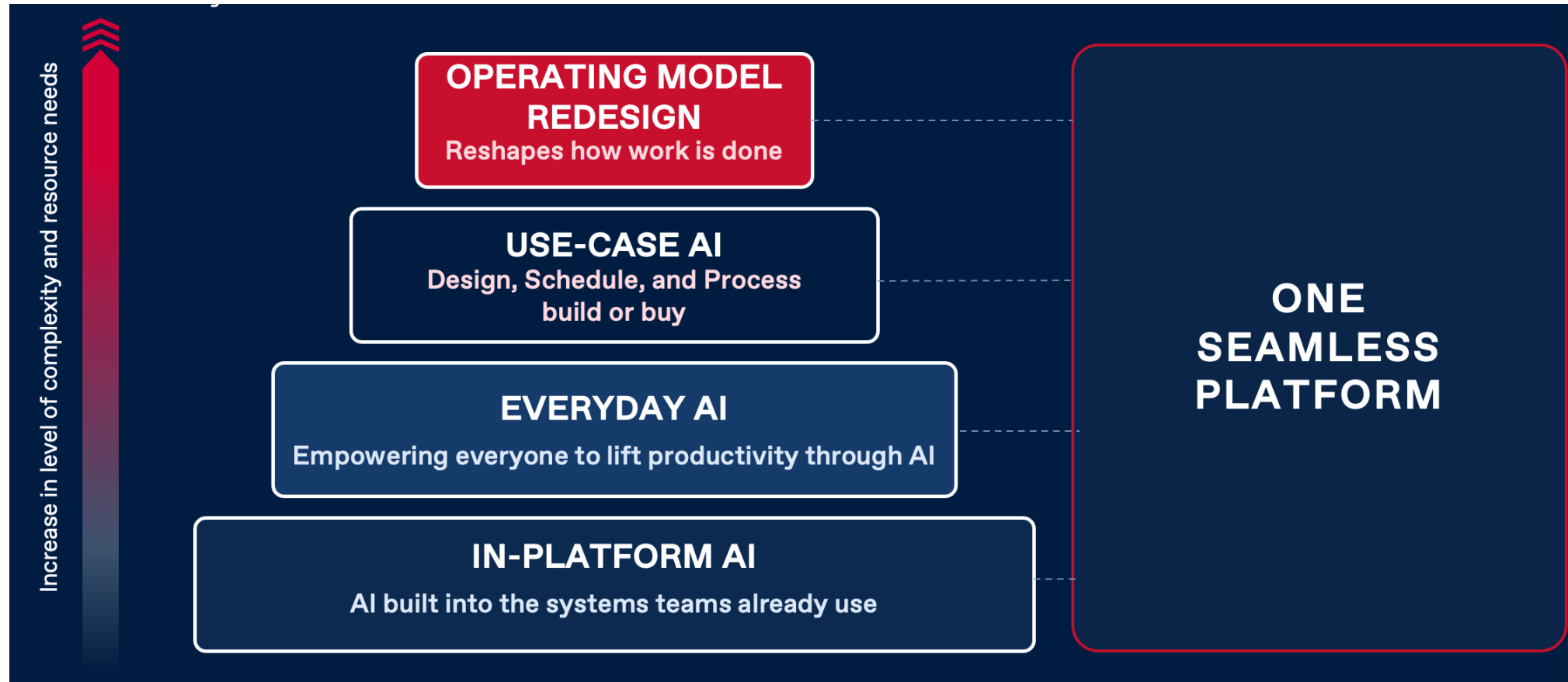
Ask me anything...

Actions

AI-generated response should be checked for accuracy.

The capability is new. The workflow is unchanged.

How we think about AI at Suffolk



Today: use-case AI + operating model redesign.

THE QUESTION

What do we do when we find an important workflow gap, and the market does not yet offer a solution that fits?

Our answer is not that we must build everything ourselves. It is that we must get much better at deciding what to buy, what to configure, and what to build.

SECTION 01

Build versus buy

A strategic allocation question, not a procurement one

ALLOCATION

Buy the substrate. Build the workflow.

BUY OR ADOPT

Standardize the plumbing

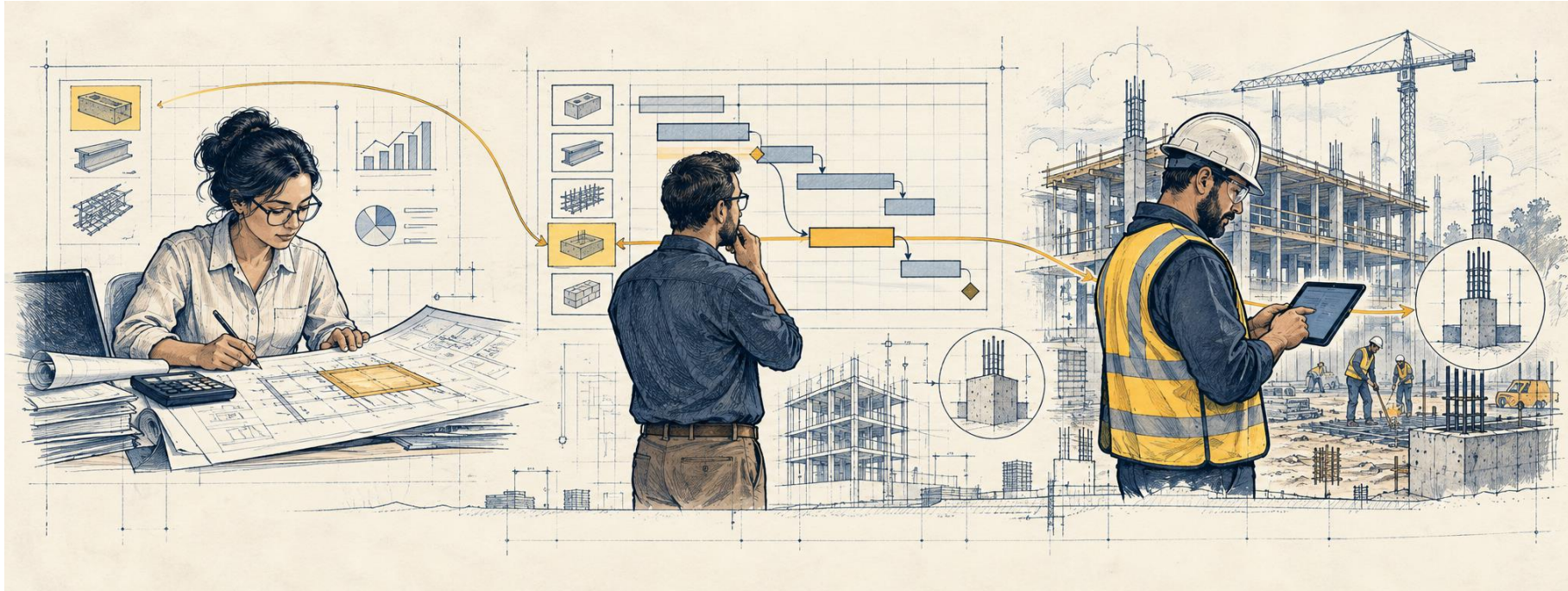
BUILD

Customize the fit

Bespoke does not mean building from scratch. It means controlling the parts that determine fit.

FIT

The same tool rarely fits two jobs



Same tool ≠ same workflow.

The workflow is the product



AI may power one step. The product spans the whole workflow.

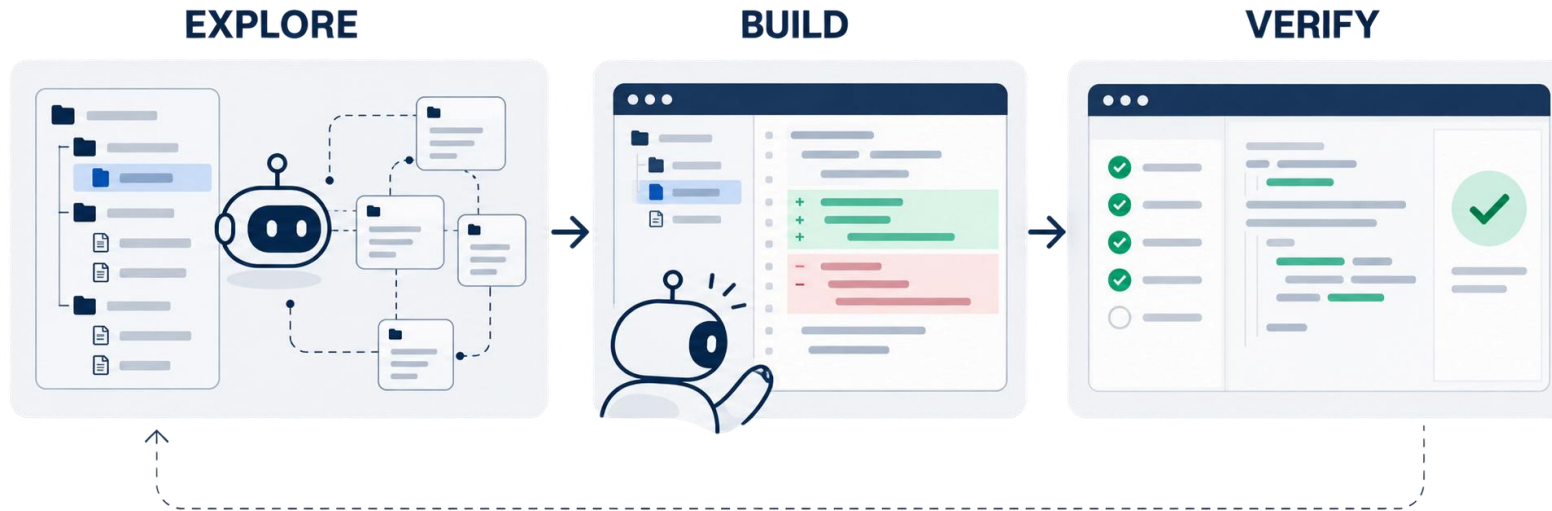
SECTION 02

Code is the immediate opportunity

It may not be an AI feature inside a product. It may be AI helping us write the product.

ACCELERATION

What AI already does well in software development



With context and guardrails, implementation gets dramatically cheaper.

The question we ask has changed

Can we afford to build it?



Can we specify it well?

Lower implementation cost changes the allocation decision.

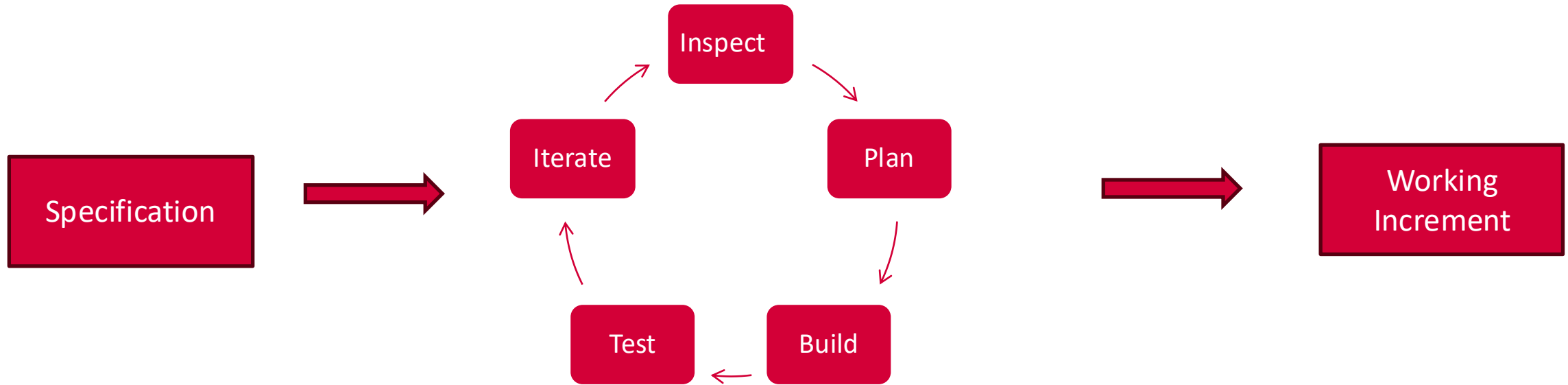
SECTION 03

Agentic coding changes the bottleneck

The agent becomes an implementation engine

MECHANISM

How the agent actually works



A specification makes the agent useful.

BOTTLENECK

The scarce resource moves upstream

CODE

gets cheaper

JUDGMENT

moves upstream

The scarce resource moves upstream.

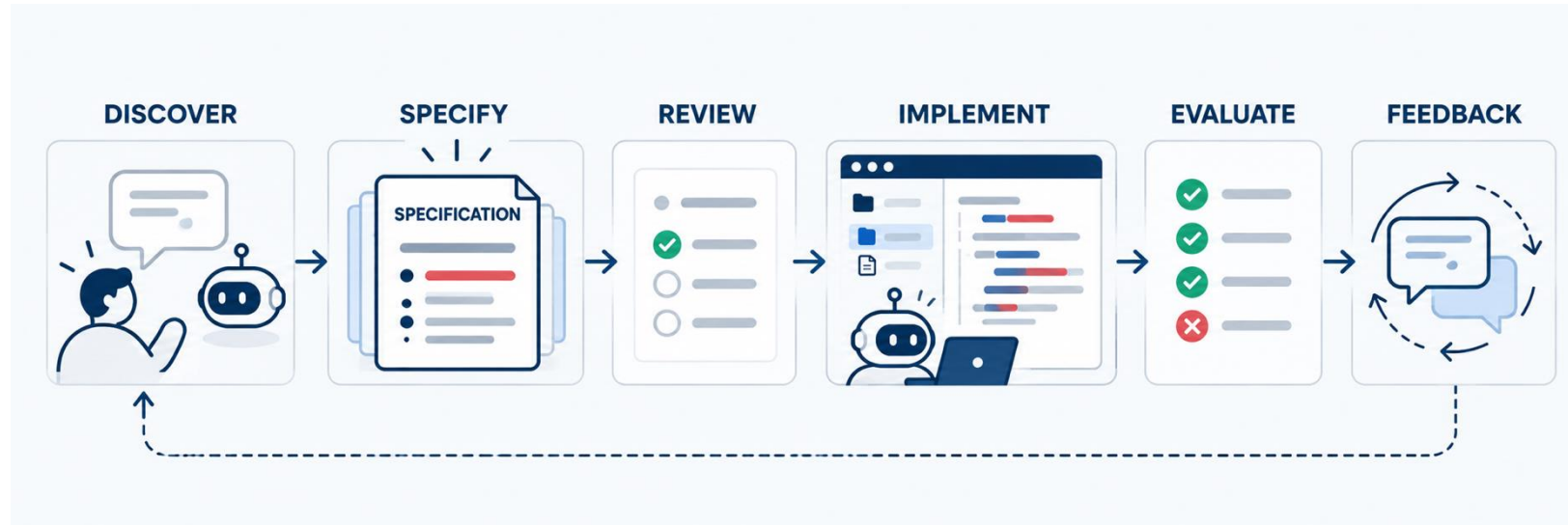
SECTION 04

Spec-driven development

The operating model that connects product intent, technical execution, and AI-assisted delivery

PROCESS

The workflow



The specification becomes the build source.

GOVERNANCE

Agile still has a place

JIRA

portfolio coordination

SPEC

build source

Less ticket ceremony, more executable clarity.

REUSE

Encoding our practice so it compounds

Skills

Packages

Standards

A pattern library without pattern lock-in.

Skill Catalog

These skills are available in every Suffolk Data project after running `setup.sh`.

Skill	When to Invoke	What It Does
using-suffolk-powers	Start of any session in a Suffolk repo	Loads skill invocation discipline and catalogs available skills
spec	Creating initiative specs, technical/data plans, or feature specs	Guides SDD workflow with templates and interview patterns
databricks-read	Reading data from Databricks in R or Python	Provides patterns for excavatoR (R) and pylon (Python) usage
writing-suffolk-skills	Creating or editing skills in suffolk-powers or initiative repos	Meta-skill for skill structure, naming, and scope decisions
eval-skill	Testing skill changes or running eval suites	Orchestrates automated skill evaluation and benchmarking
suffolk-logging	Writing or modifying R/Python logging, timing, or log-level thresholds	Defines what to log, at what level, and what to avoid (excavatoR / pylon)
llm-calls	Making LLM calls from Suffolk R or Python code	Routes to approved wrappers (<code>excavatoR::llm_chat</code> / <code>pylon.llm_chat</code>) and covers prompt handling, logging, chunking, and structured output
llm-evaluations	Evaluating or regression-testing LLM outputs in an SDD workflow	Builds eval cases from a spec and routes to the approved grading engine
suffolk-posit-app-development	Developing or reviewing apps in the <code>suffolk-posit-projects</code> monorepo	Covers monorepo rules, language standards, libraries, deployment, and testing

REUSE

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A pattern library without pattern lock-in.

excavatoR

License: **GPLv3** R: **>=3.6.0** coverage: **61%** Run R tests (excavatorR): **passing**

Common Functions to Help Conduct Analysis

excavator is Suffolk Construction's R package for standardizing common functions and workflows used in data analysis, following the Suffolk Data Analysis Common Analytics Framework. This package provides utilities for data visualization, Databricks integration, presentation generation, and AI-powered analysis.

Features

Data Visualization

- **Suffolk Brand Themes:** Custom ggplot2 themes that follow Suffolk's brand guidelines
- **Color Palettes:** Standardized color schemes for consistent visualizations
- **Chart Standardization:** Consistent styling across all analytical outputs

Brand Assets

- **Brand YAML from Connect:** Fetch and parse `_brand.yml` from Posit Connect, with automatic asset downloading and Shiny resource path registration
- **Direct Logo Access:** Logo paths are rewritten to servable URLs — use `brand$logo$image$primary` directly in `tags$img(src = ...)`
- **Font Awesome icons:** Self-hosted Font Awesome icons served from Posit Connect — drop icons into Shiny apps with a single `read.icon()` call, no manual CSS setup required

Table Branding

- **Branded gt Tables:** Apply Sufolk brand styling to `gt` tables with `brand_table(gt())`. Supports two styles:
 - `style = "standard"` (default): refined house style with navy header, red underline, white body, zebra striping, and grid lines. All colors resolve from the brand object.
 - `style = "minimal"`: baseline `brand_gt` colors plus font embedding only.
- **Self-Contained Fonts:** Brand fonts are base64-embedded via `@font-face` CSS so HTML output works offline.
- **Reusable Font CSS:** `brand_font_css()` generates `@font-face` CSS with base64-encoded font data from the brand, reusable across `gt`, Quarto, and any HTML output.

Note on R vs Python parity: The R side adopts Posit's `brand.yml::theme_brand_gt()` directly. The Python equivalent (`great_tables`) does not yet consume `brand.yml` and is tracked separately.

Databricks Integration

- **Seamless Connection:** Easy connection to Databricks clusters
- **Secret Management:** Secure handling of Databricks secrets and authentication
- **Data Pipeline Support:** Utilities for data extraction and manipulation

Presentation Tools

- **PowerPoint Generation:** Create presentations using Suffolk-branded Quarto templates
- **Quarto Brand Materialization:** Download the Connect-hosted `_brand.yml` and reveals extension into a project so Quarto can render branded reports and presentations
- **Automated Reporting:** Streamlined workflow from analysis to presentation

AI & Machine Learning

- **LLM Chat Interface:** Integration with multiple AI providers (Databricks, OpenAI, Anthropic, AWS, Azure)
- **LLM Evaluation Framework:** Systematic evaluation of LLM performance on question-answering tasks
- **Prediction Parsing:** Utilities for handling machine learning model outputs
- **Data Quality:** Outlier detection and data validation tools

A package to assist Suffolk Data Analysts and Data Scientists with common data-related tasks. See [excavator](#) for R version.

 Badges

coverage 88% Run tests passing

Table of Contents

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- [Features](#)
- [Usage](#)
- [Development](#)
- [License](#)

⚡ Installation

This package is intended for internal use only.

To install locally:

```
git clone https://github.com/Suffolk-Construction-Corporation-Inc/pylon.git
cd pylon
pip install -e .
```

To also generate `manifest.json` for Posit Connect deployment (see `make_quarto` below), install the optional `deploy` extra:

```
pip install -e ".[deploy]"
```

Features

- **connect_to_databricks :**
Seamlessly connect to Databricks warehouses for easy access to Suffolk's enterprise data.
- **databricks_host :**
Resolve a Databricks workspace hostname for a given environment name (e.g. "prod", "test") without relying on environment variables.
- **databricks_token :**
Obtain a raw Databricks bearer token for the current runtime. On Posit Connect the viewer's OAuth credentials are used; on Workbench managed credentials are used; locally PAT/CLI/env-var fallbacks apply.
- **lia_utils :**
Seamlessly use chatlas-based LLM calls without the hassle of explicit setup.
- **logging_utils :**
Handle several different logging functions that are pre-formatted to maintain consistency.
- **suffolk_color_palette :**
Generate Suffolk-branded color palettes for consistent data visualizations.
- **read_brand_yaml :**
Fetch and parse a `_brand.yml` file from Posit Connect, download associated font and image assets, and return the parsed brand configuration for use in Shiny apps. Mirrors `excavatorR::read_brand_yaml()`.
- **read_brand_local :**
Parse a `_brand.yml` already materialized on disk, with no Connect fetch. Used by the `_brand` and Quarto project scaffolds with `make_quarto()`, since `pre-render.py` has already downloaded the brand; avoids a second full brand fetch (roughly 88 MB) on every render.

REUSE

Encoding our practice so it compounds

Skills

Packages

Standards

Ops Finance wants a dashboard combining all subcontractor co

Idea Evaluation

Problem

What is the problem and why does it matter? What happens if v

Opportunity & Value

How could we improve this? What value or capability is unlocke

Desirability (Business Demand)

How strong is business demand? Who is asking for this and wh

Feasibility / Readiness

Test	Response
Data Readiness Does systemized, reliable data exist?	
Process Readiness Is there a stable process this fits into?	
User Readiness Can the business adopt this?	
Technical Readiness Can we build this (skill/ tech?)	
AI Readiness Can AI (Devin, etc.) be leveraged?	
Scalability Potential Can this solve other use cases?	

Initiative Brief

Strategic Context

What is happening in the business today, and why does this matter nov

Decision & Capability Gap

Decision to Improve

What decision or workflow are we trying to improve?

Current Gap

- What is not working today?
- Where are the breakdowns (data, process, consistency, timing)?

Target Capability (Long-Term Vision)

What capability should exist if this is fully realized?

- What new behavior or workflow does this enable?
- How should the business operate differently?

Why Now (Priority)

Why should we invest in this now vs later?

- What makes this urgent or important today?
- What happens if we delay?

1. Overview

Field

Value

Product	
Technical Product Manager	
Data Strategy & Ops	
Executive Sponsor	
Team	

Description\

{{Brief description of the product. What it does, who it serves, and the core value proposition.}}

2. Problem Statement

{{Describe the core problem this product is solving.}}

This leads to: - {{Pain point 1}}\

- {{Pain point 2}}\
- {{Pain point 3}}

3. Goals & Non-Goals

Goals

- {{Goal 1}}\
- {{Goal 2}}\
- {{Goal 3}}

Non-Goals

- {{Explicitly out-of-scope item 1}}\
- {{Explicitly out-of-scope item 2}}\
- {{Explicitly out-of-scope item 3}}

A pattern library without pattern lock-in.

SECTION 05

The infrastructure we do not have to build

Agentic coding makes development faster. It does not eliminate the need for a platform.

The platform removes the infrastructure tax

Posit Connect



Posit Connect

Shiny Dashboards | Quarto Reports |
Vetiver APIs | Databricks OAuth

Posit Workbench



Posit Workbench

R & Py environments | Version Control |
Feature Eng | Model Training

Databricks



Databricks

Unity Catalog | Delta Tables |
Metric Views | API Gateway

One body of code can move from discovery to production.

DIVISION OF LABOR

What low DevOps overhead actually means

**WE OWN
FIT**

**THE PLATFORM ABSORBS
PLUMBING**

A data product team can work like a software team without becoming an infrastructure company.

SECTION 06

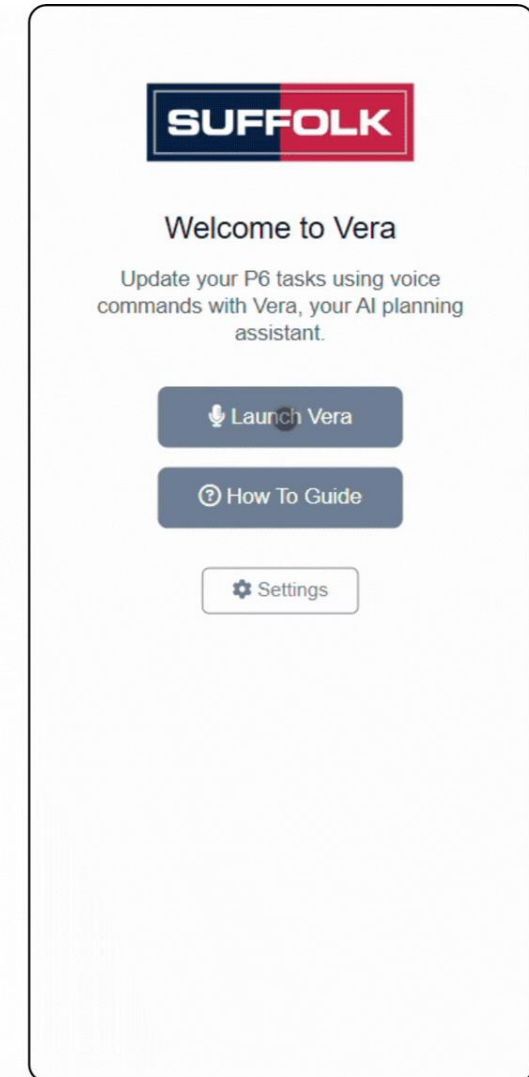
From code to jobsite

Bespoke applications in practice

Updating the schedule from the field

VOICE → REVIEW → WRITE-BACK

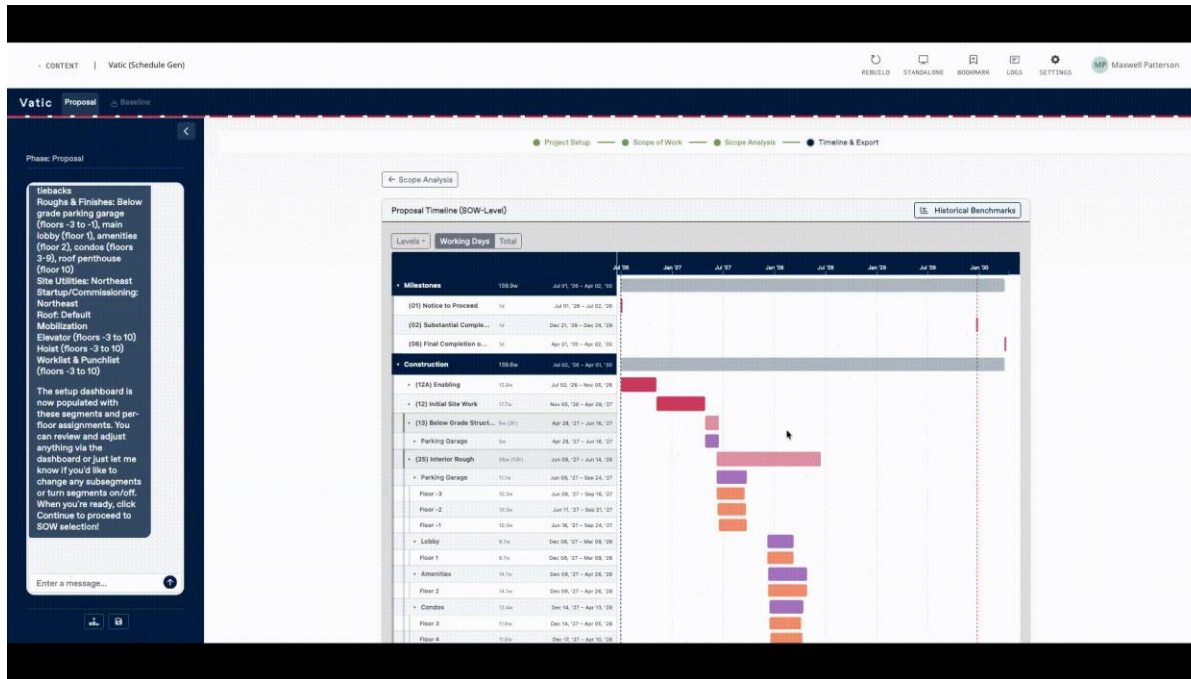
The value is the workflow around the model.



USE CASES

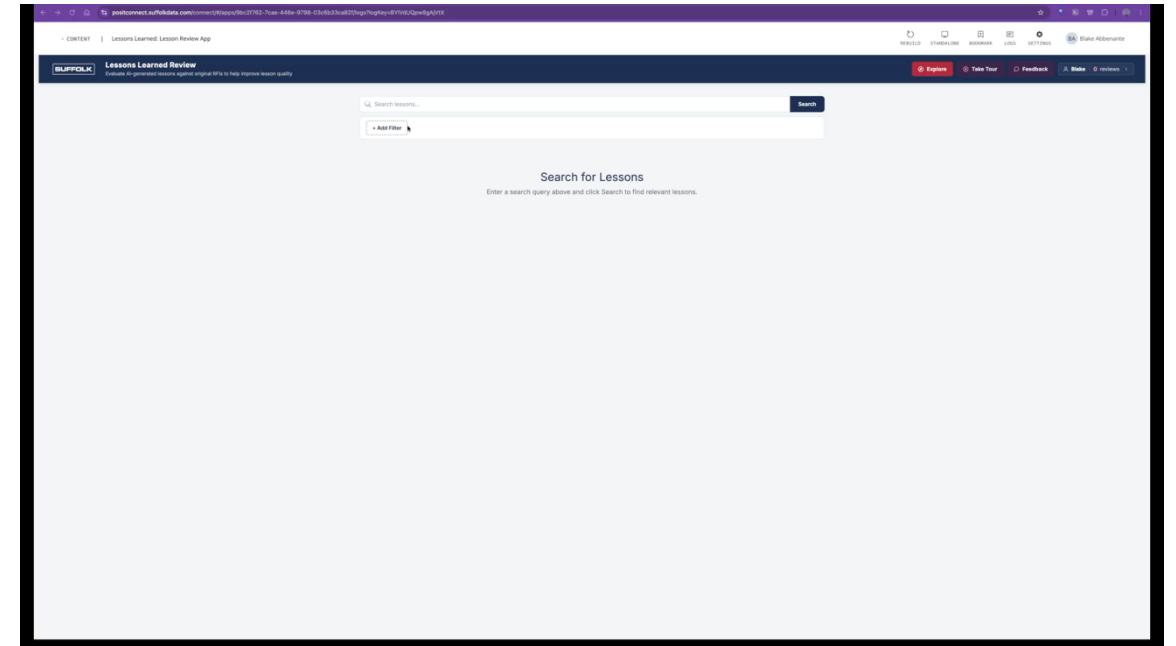
Two more bespoke applications

GENERATIVE SCHEDULING



A strong first draft for planners

LESSONS LEARNED



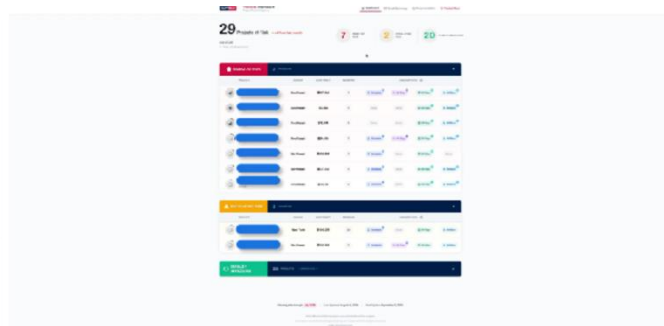
Grounded retrieval from project history

Neither replaces expertise. Both shorten the distance to a good decision.

USE CASES

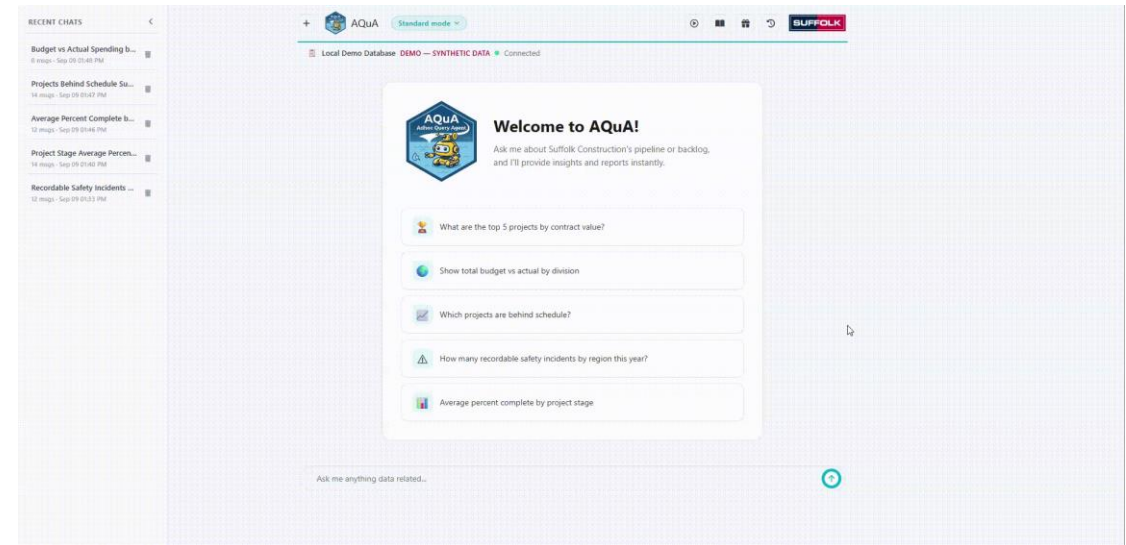
The pattern repeats across the portfolio

TRIPWIRES



Prioritized risks and exceptions

AQuA



Grounded questions against curated data

Different problems produce different products. The underlying model stays consistent.

THE MODEL

Underneath every one of them, the same six things

01

Workflow

02

App

03

Data

04

Process

05

Delivery

06

Judgment

Code-first gives us flexibility.

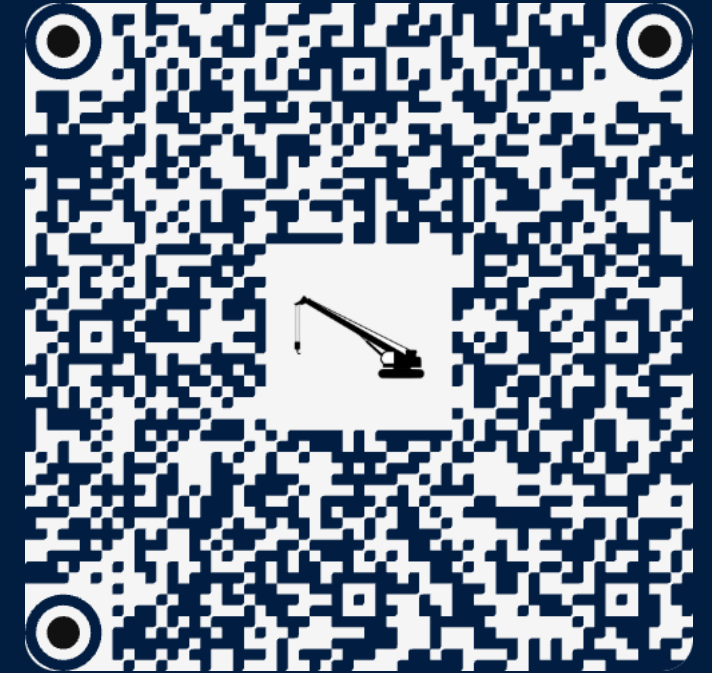
Agents give us acceleration.

Managed infrastructure gives us leverage.

The near-term advantage comes from using AI to build the construction software we still need.

Thank you

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@gottalottasoul



@abbenante

